

CStats - Lab #8

April 28, 2026

Exercise 1.

Let the cost function be $f(x) = x^2 - 4x + 4$.

(a) Find the global minimum analytically.

(b) Perform two steps of Gradient Descent by hand. Use an initial starting point of $x_0 = 0$ and the learning rate $\eta = 0.1$.

Solution 1.

(a) We set the derivative to zero, $f'(x) = 0$, to find the global minimum. Thus, we have

$$f'(x) = 2x - 4 \rightarrow 2x - 4 = 0 \rightarrow x = 2. \quad (1)$$

(b)

- Calculate gradient at $x_0 = 0$, which gives us $f'(x_0) = f'(0) = -4$.
- Using the update rule $x_1 = x_0 - \eta f'(x_0)$, we obtain $x_1 = 0.4$.
- Calculate gradient at $x_1 = 0.4$, $f'(x_1) = -3.2$.
- Update $x_2 = x_1 - \eta f'(x_1) = 0.72$.

Then, let's check whether the algorithm is moving in the right direction. Yes, $0 \rightarrow 0.4 \rightarrow 0.72$, It's moving toward 2 which is the global minimum.

Exercise 2.

We want to fit a line without a bias term $y = wx$ to a dataset with two data points:

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 4)$$

We use the Mean Squared Error as the loss function for a single data point $L_i(w) = \frac{1}{2}(\hat{y}_i - y_i)^2 = \frac{1}{2}(wx_i - y_i)^2$.

(a) What is the derivative of the loss function with respect to w for a single data point i ?

(b) Start with $w_0 = 0$ and a learning rate $\eta = 0.1$. Perform one update using data point 1, and then a second update using data point 2.

Solution 2.

(a) Using the chain rule, we get the derivative of the loss function as $\frac{\partial L_i}{\partial w} = (wx_i - y_i)x_i$.
 (b) We write the following steps to get the Stochastic Gradient Descent updates:

- Using (x_1, y_1) , $(w_0x_1 - y_1)x_1 = (0 - 2)1 = -2$.
- Update: $w_1 = w_0 - \eta \frac{\partial L_i}{\partial w} = 0.2$.
- Using (x_2, y_2) , $(w_1x_2 - y_2)x_2 = -7.2$.
- Update: $w_2 = w_1 - \eta \frac{\partial L_i}{\partial w} = 0.92$.

Using the given data points, if we solve for w , we obtain $w = 2$. Using the gradient descent or stochastic gradient descent, we see how the algorithm mathematically hunts for this number starting from 0. One full pass through the dataset is called one epoch. After one epoch, the gradient descent reaches $w = 0.5$ and the stochastic gradient descent reaches $w = 0.92$. Even though both algorithms looked at the exact same amount of data, stochastic gradient descent got much closer to the true weight since stochastic gradient descent used the information from data point 1 to improve itself before it even looked at data point 2.

Exercise 3.

We have an input x , one hidden layers, and one output layer such that

$$a_1 = w_1x + b_1$$

$$h_1 = \sigma(a_1)$$

$$\hat{y} = w_2h_1 + b_2$$

The loss function is $L = \frac{1}{2}(\hat{y} - y)^2$ Given a single training example (x, y) , derive the update rules for w_1, b_1, w_2, b_2 using a learning rate η .

Solution 3.

Using the chain rule,

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_2} \quad (2)$$

We have $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$ and $\frac{\partial \hat{y}}{\partial w_2} = h_1$. Similarly, for b_2 ,

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b_2} = (\hat{y} - y)1. \quad (3)$$

Thus, the update rules are $w_2 = w_2 - \eta(\hat{y} - y)h_1$ and $b_2 = b_2 - \eta(\hat{y} - y)$.

For w_1 , we write the following:

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_1} \frac{\partial h_1}{\partial a_1} \frac{\partial a_1}{\partial w_1} = (\hat{y} - y)w_2\sigma'(a_1)x \quad (4)$$

Finally, for b_1 , we have

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_1} \frac{\partial h_1}{\partial a_1} \frac{\partial a_1}{\partial b_1} = (\hat{y} - y)w_2\sigma'(a_1)1 \quad (5)$$

Thus, the update rules are $w_1 = w_1 - \eta(\hat{y} - y)w_2\sigma'(a_1)x$ and $b_1 = b_1 - \eta(\hat{y} - y)w_2\sigma'(a_1)$.

Exercise 4.

Suppose you build a Neural Network with 10 hidden layers, but you do not use any activation functions (like ReLU, Sigmoid, Softmax, etc.). Mathematically, what does this 10-layer network simplify to, and why is this a problem for complex datasets?

Solution 4.

A layer without an activation function is just a linear transformation: $Z = WX + b$. If we stack two layers: $Z_2 = W_2(W_1X + b_1) + b_2$. Expanding this: $Z_2 = (W_2W_1)X + (W_2b_1 + b_2)$. Since a product of two matrices is just another matrix (W_{new}) and the bias terms simplify to a single vector (b_{new}), the entire 10-layer network collapses into a single linear model:

$$Y = W_{final}X + b_{final}$$

As a result, linear models can only find linear decision boundaries and cannot learn the complex patterns.

Exercise 5.

Imagine you are training a model on the large dataset st. 100,000 data points. Explain why calculating the gradient using standard Gradient Descent might be problematic and how does Stochastic Gradient Descent solve this?

Solution 5.

Gradient Descent requires loading all 100,000 data points and performing a massive matrix multiplication just to take one single step. This is slow and often exceeds RAM/GPU memory if we use a standard computer. Stochastic Gradient Descent takes a step after looking at only a small fraction of the data, which allows the model to start learning immediately.